



DEPARTMENT OF CHEMICAL ENGINEERING NATIONAL INSTITUTE OF TECHNOLOGY SRINAGAR

Lecture-3 Bioresource Technology

BIOMASS

➤ INTRODUCTION

- AN IMPORTANT BIORESOURCE
- REFERS TO ALL ORGANIC MATTER GENERATED THROUGH THE PROCESS OF PHOTOSYNTHESIS
- ALSO INCLUDES MATTER PRODUCED BY LAND AND WATER BASED VEGETATIONS
- RENEWABLE AND BIODEGRADABLE SOURCE OF ENERGY
- ULTIMATE SOURCE OF ENERGY IS THE SUN
- PLANTS/VEGETATIONS CAPTURE THIS ENERGY DURING PHOTO SYNTHESIS

➤ EXAMPLES OF BIOMASS

- Biomass includes agricultural and agro-industrial wastes and residues, energy plantation, farm wastes, forest products, and wastes, animal waste matter, aquatic biomass etc.
- Indian context: animal wastes and cattle dung have been considered as separate category and not as a part of biomass.

➤ BIOMASS SYNTHESIS IN NATURE (FIGURES: 1-3)

- Biomass is composed of cellulose, hemicellulose, lignin and extractives
- Organic matters that form the part of biomass are produced by reaction between CO_2 in the air, water and sunlight during photosynthesis
- Less than 1% of the available sunlight is stored as chemical energy by the plants
- When the carbon hydrogen and oxygen bonds i.e. C-C, H-H, C=O,..... are broken by digestion, combustion, or by decomposition, stored chemical energy gets released.

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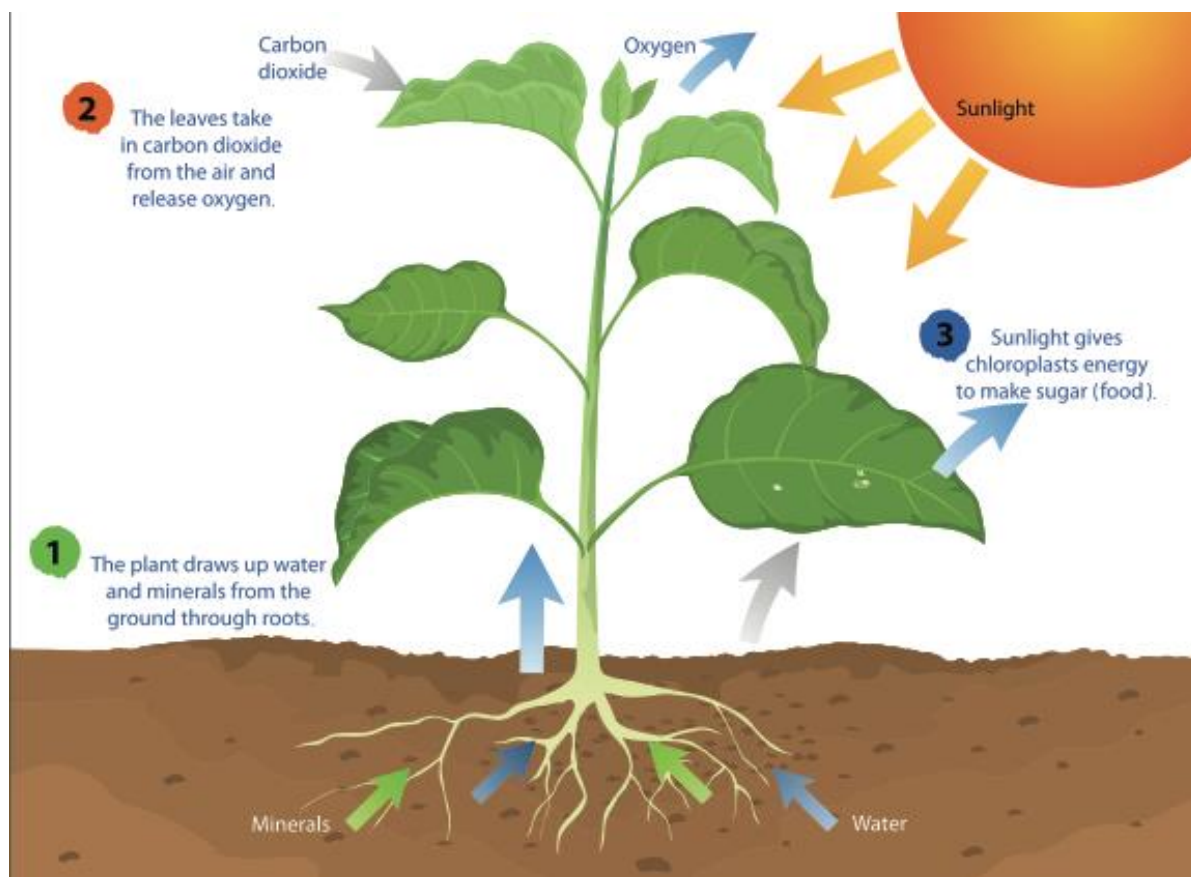


Figure-1

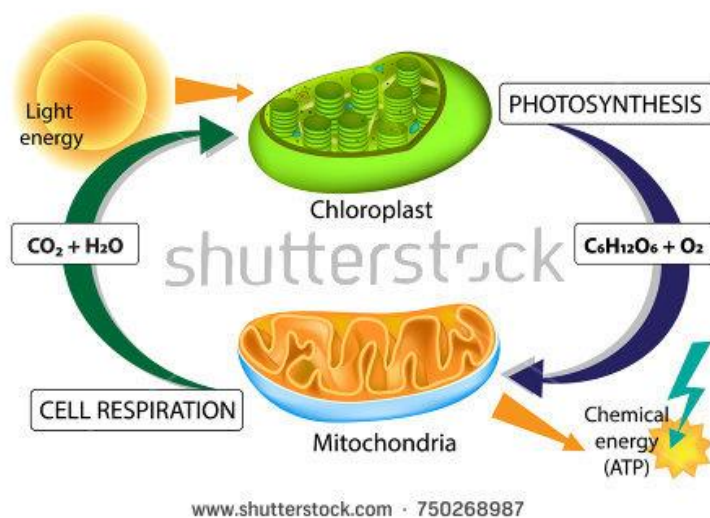


Figure-2

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Example of a photosynthesis Flow chart

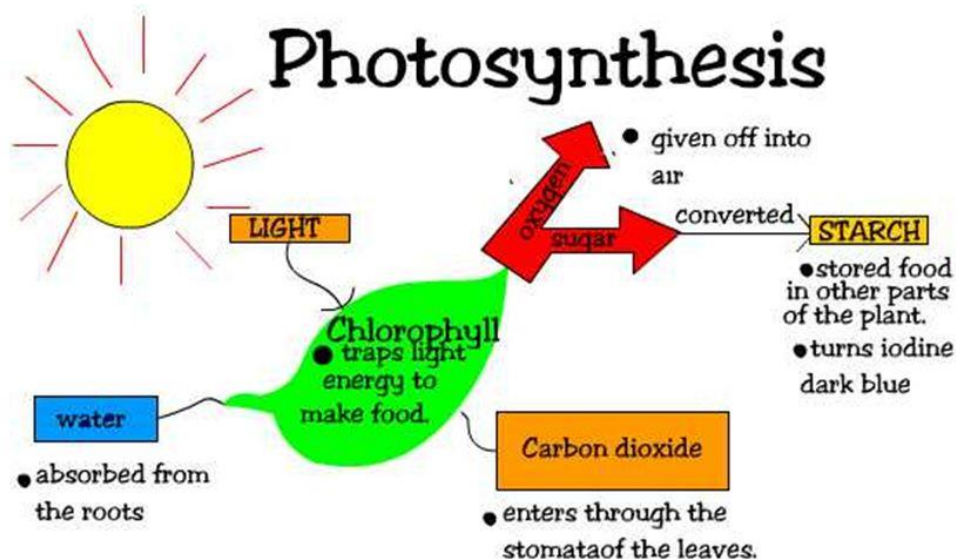


Figure-3

➤ SIGNIFICANCE OF BIOMASS

- Continuous depletion of fossil fuels, rising prices of petroleum, increasing demand for energy and environmental concerns are the critical reasons due to which there is emphasis on alternate renewable bioresources, like biomass
- Biodegradability, environmental friendliness and sustainability are the important features which have made the biomass, a primary candidate.
- It is the most abundant renewable alternate energy source evenly distributed in the world.
- Fossil fuels can be substituted by biomass in power plants.
- Biomass can be interpreted as 'stored solar energy'.
- It is benign in terms of carbon sequestration.
- It is non-toxic and carbon neutral source.



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- It has low amount of ash, nitrogen and sulfur contents which have made the biomass more attractive from the stand point of environment.

➤ WORLD ENERGY SCENARIO

- Due to the population growth and industrial development, demand for energy is increasing day by day.
- Bioresources, like biomass is highly significant for meeting the challenges along with other alternate sources, such as wind, solar, tidal and geothermal energies.
- Fast growing economies of China and India accounts for over half of the increase.
- China is believed to be the largest market whereas consumption in OECD (Organization of Economic Cooperation and Development) is slow as reported by British Petroleum Energy Outlook 2018.
- By the year 2040, it is believed that the demand would be 18 billions of oil equivalent (toe), **Figure-4**.
- **Figure-5** shows that renewable and nuclear powers are the rapidly-expanding sources of energy.

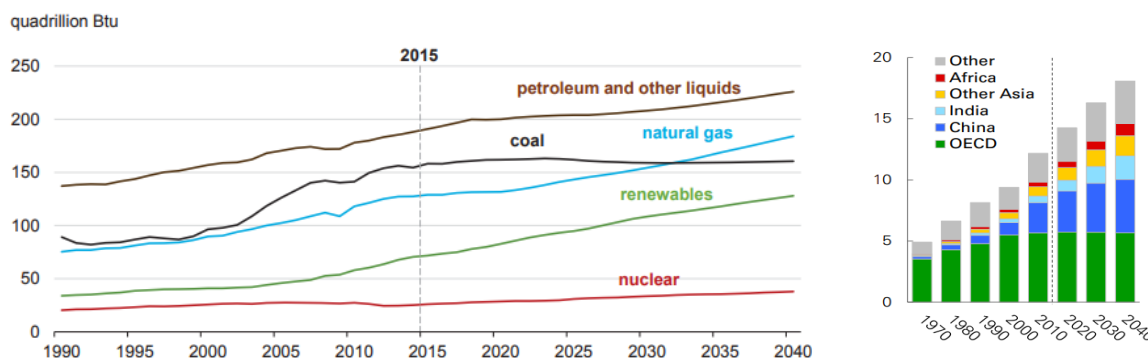


Figure-4: Global energy demand (BP Energy Outlook, 2018).



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**Figure-5: World energy consumption by different types of energy sources.
(International Energy Outlook, 2018)**

- Renewable energy sources are rising by an average of 2.8%, natural gas 2.1%, nuclear generation 1.5% per year, whereas coal contribution drops from 40% to 31% over a period of 2015 to 2040.
- For a country like India, the use and exploitation of renewable resources (including bioresources) has become more challenging.
- Biomass occupies unique place among the various alternate renewable energy sources.
- India has a high energy target of 175 GW by 2022 and 275 GW by 2026-2027.
- India has decided to achieve 40 GW from solar roof tops by 2022 (WEC, 2018)
- Rest 135 GW would be generated through other sources, where biomass plays significant role.

➤ **PHYSICO-CHEMICAL PROPERTIES OF BIOMASS MATERIAL**

- Very significant from the stand point of conversion technology
- Effect the cost of conversion.
- Conversion of biomass to energy and other products (chemicals) are effected by characteristics of biomass.



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- The properties of interests are: **moisture contents, heating values, fixed carbon, volatile matter, ash content, alkali metal, cellulose/lignin ratio and elemental composition.**

► Moisture Content

- It is the water content stored in spaces within the dead cells and the cell walls.
- When the fuel is dried, the stored moisture equilibrates with the ambient relative humidity.
- The moisture present in biomass comprises intrinsic moisture and extrinsic moisture.
- Intrinsic moisture content of the material is not influenced by the weather whereas the extrinsic moisture is influenced by the prevailing weather conditions.
- Moisture contents of some typical biomass after ambient drying have been reflected in **Table-1**.

Table-1: Proximate and Ultimate Analysis of some Common Biomass Feed Stocks (wt. %)

Biomass	Proximate analysis				Ultimate analysis ^{daf}					References
	M	VM ^{db}	FC ^{db}	A ^{db}	C	H	N	S	O	
Woody Biomass										
Pine chips	7.6	72.4	21.6	6	52.8	6.1	0.5	0.09	40.5	Masia (2007)
Poplar	6.8	85.6	12.3	2.1	51.6	6.1	0.6	0.02	41.7	Miles et al. (1995)
Sawdust	34.9	84.6	14.3	1.1	49.8	6	0.5	0.02	43.7	Tillman (2000)
Willow	10.1	82.5	15.9	1.6	49.8	6.1	0.6	0.06	43.4	Moilanen (2006)
Grasses and Herbs										
Herbaceous, Agricultural Biomass	12	75.2	19.1	5.7	49.9	6.2	1.2	0.15	42.6	Vassilev (2010)
Switch grass	11.9	80.4	14.5	5.1	49.7	6.1	0.7	0.11	43.4	Miles et al. (1995)
Sorghastrum grass	11.3	81.6	14.2	4.2	49.4	6.3	0.3	0.05	44	Miles et al. (1995)
Straws										
Rice straw	76	64.3	15.6	20.1	5.1	5.7	1.0	0.16	43	Miles et al. (1995), Thy et al. (2006)



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Wheat straw	10.1	74.8	18.1	7.1	49.4	6.1	0.7	0.17	43.6	Vassilev (2010)
Aquatic Biomass										
Marine Macroalgae	10.7	50.5	25.9	23.6	43.2	6.2	2.2	2.6	45.8	Tite (2006)
Duckweed	14	78	8.8	26	30	4.3	2.1	0.8	21	Duan (2013)

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db: dry basis, daf: dry ash free basis

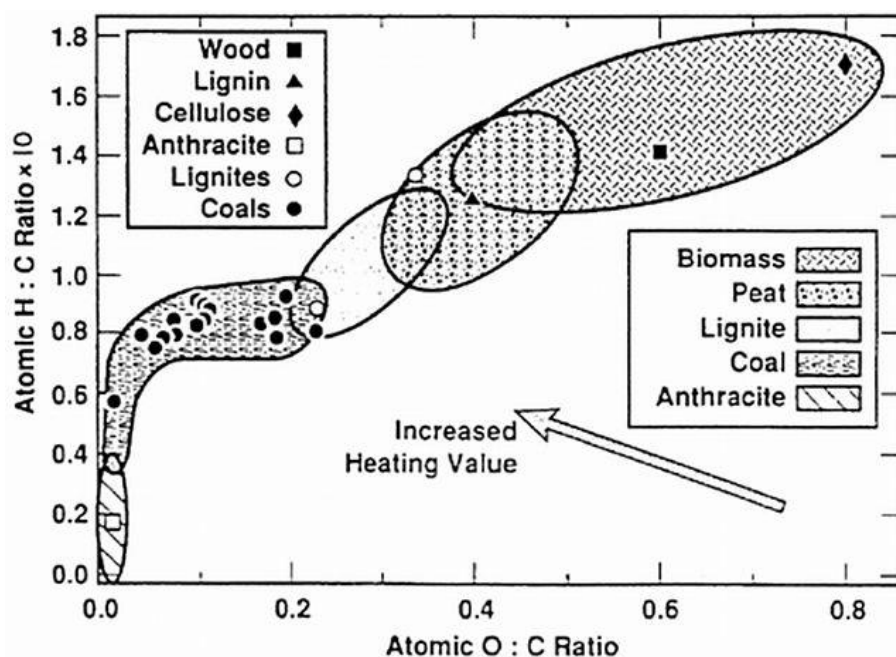
- For conversion, like anaerobic digestion, the feedstock with only high moisture content can be effectively processed.
- Whereas thermochemical conversion processes require a biomass with low moisture content.
- Water in biomass is not a combustible material and has a negative impact on the overall energy balance.
- However, some moisture content is required in the gasification process of the biomass for the production of hydrogen which is increased with moisture.
- Moisture in biomass decreases its heating value.
- The presence of water in biomass influences its behavior during pyrolysis and affects the physical properties and the quality of the pyrolysis products.

➡ Proportions of Fixed Carbon/Volatiles

- For solid fuels chemical energy is stored in two forms, fixed carbon and volatiles.
- The fixed carbon content (FC) is the mass remaining after the releases of volatiles excluding the ash and moisture contents.

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- Whereas the volatiles content, or volatile matter (VM) of a solid fuel is that portion driven-off as a gas (including moisture) by heating (to 950 °C for 7 min).
- Proximate analysis of a fuel is based upon the determination of VM content, ash, moisture and the FC (determined by difference).
- Standard laboratory tests are used to determine the proximate analysis of the fuel. **Table-1** gives the proximate analyses of some typical biomass sources.
- Elemental analysis of a fuel, presented as C, N, H, O and S together with the ash content is termed as the ultimate analysis of a fuel.
- **Table-1** gives the ultimate analysis data for some biomass materials.
- The significance of the VM and FC contents is that they provide a measure of the ease with which the biomass can be ignited and subsequently gasified, or oxidized, depending on how the biomass is to be utilized as an energy source.
- The significance of the O/C and H/C ratios on the HV of solid fuels can be illustrated using a Van Krevelen diagram shown in **Figure-6**.
- Comparison of biofuels with fossil fuels, such as coal, shows clearly that the higher proportion of oxygen and hydrogen, compared with carbon, reduces the energy value of a fuel, due to the lower energy contained in carbon–oxygen and carbon–hydrogen bonds, than in carbon–carbon bonds.





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Figure-6: Van Krevelen diagram.

► Heating Value

- The heating value (HV) of a material is an expression of the energy content, or heat value, released when burnt in air.
- The HV is usually measured in terms of the energy content per unit mass, or volume.
- The HV of a fuel can be expressed in two forms, the gross HV (GHV), or higher heating value (HHV) and the net HV (NHV), or lower heating value (LHV).
- The HHV is the total energy content released when the fuel is burnt in air, including the latent heat contained in the water vapour and, therefore, represents the maximum amount of energy potentially recoverable from a given biomass source.
- In practical terms, the latent heat contained in the water vapour cannot be used effectively and, therefore, the LHV is the appropriate value for the subsequent use of the energy available in biomass.
- Based on proximate analysis many correlations have been developed from time to time by various researchers relating HHV in terms of FC and VM of the biomass as detailed below in **Table-2**:

Table-2: Correlations for HHV based on Proximate Analysis



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S. No.	Correlation	Reference
1	$\text{HHV (MJ/kg)} = 0.196 (\text{FC}) + 14.119$	Demirbas (1997)
2	$\text{HHV (kJ/kg)} = 354.3\text{FC} + 170.8\text{VM}$	Cordero et al. (2001)
3	$\text{HHV (MJ/kg)} = 0.3536\text{FC} + 0.1559\text{VM} - 0.0078\text{Ash}$	Parikh et al. (2005)
4	$\text{HHV (MJ/kg)} = 0.1905\text{VM} + 0.2521\text{FC}$	Yin (2011)

- Based on ultimate analysis too, a number of correlations have been developed for HHV as function of elemental carbon, hydrogen, oxygen, sulfur and nitrogen.
- Only a few of these are given below in **Table-3**:

Table-3: Correlations for HHV based on Ultimate Analysis

S. No.	Correlations	References
1	$\text{HHV (MJ/kg)} = 0.4373\text{C} - 1.6701$	Tillman (1978)
2	$\text{HHV (MJ/kg)} = 0.3491\text{C} + 1.1783\text{H} + 0.1005\text{S} - 0.1034\text{O} - 0.0151\text{N} - 0.0211\text{Ash}$	Channiwala and Parikh (2002)
3	$\text{HHV (kJ/kg)} = 3.55\text{C}^2 - 232\text{C} - 2230\text{H} + 51.2\text{C} \times \text{H} + 131\text{N} + 20600$	Friedl et al. (2005)
4	$\text{HHV} = 0.2949\text{C} + 0.8250\text{H}$	Yin (2011)

➡ **Ash/Residue Content**

- The chemical breakdown of a biomass fuel by either thermochemical or bio-chemical processes produces a solid residue.
- When produced by combustion in air, this solid residue is called 'ash'.
- It is the inorganic residue that remains after combustion of the fuel in air at specific temperature.



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- It is the measure of the non-combustible materials which may contain the oxides of Si, Fe, K, Mg, Ca, Al, etc.
- Ash acts as a heat sink in the same way as moisture, lowering the combustion efficiency.
- Soluble ionic compounds in ash can have a catalytic effect on the pyrolysis and combustion of the fuel.
- The composition of mineral matter can vary between and within each biomass sample.
- The ash content of biomass affects both the handling and processing costs of the overall biomass energy conversion cost.
- During biochemical conversion, the percentage of solid residue will be greater than the ash content formed during combustion of the same material.
- For biochemical conversion process, the solid residue represents the quantity of non-biodegradable carbon present in the biomass.
- This residue will be greater than the ash content because it represents the recalcitrant carbon which cannot be degraded further biologically but which could be burnt during thermochemical conversion.
- Depending on the magnitude of the ash content, the available energy of the fuel is reduced proportionately.
- In a thermochemical conversion process, the chemical composition of the ash can present significant operational problems.
- This is especially true for combustion processes, where the ash can react to form a 'slag', a liquid phase formed at elevated



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temperatures, which can reduce plant throughput and result in increased operating costs.

➡ **Alkali Metal Content**

- The alkali metal content of biomass i.e. Na, K, Mg, P and Ca is especially important for any thermochemical conversion processes.
- The reaction of alkali metals with silica present in the ash produces a sticky mobile liquid phase which can lead to blockages of airways in the furnace in power plants.

➡ **Cellulose/Lignin Ratio**

- The proportions of cellulose and lignin in biomass are important in energy conversion processes.
- The biodegradability of cellulose is greater than that of lignin.
- Hence the overall conversion of the carbon-containing plant material present as cellulose is greater than that for plants with a higher proportion of lignin.
- It is a determining factor while selecting biomass plant species for biochemical processing.
- HHV of biomass fuels increases with increasing lignin contents.